Lesson 1: Paper Power

Year 1/2 STEAM | Unit: Stronger Together: Paper Makers | CRT plan | 45 minutes

No preparation needed. Everything is handed out by student helpers during the lesson. **No screens** are used.

This week each pair tests three paper shapes, and each student glues a record sheet into their **STEAM journal**. Journals are on the shelves as you walk in the door.

Learning Intention and Success Criteria

Learning Intention:

To discover that changing the shape of paper can change how strong it is.

Success Criteria:

I can:

- fold and roll paper into different shapes.
- test how many pebbles each shape can hold.
- say which shape was strongest and why.

Read these aloud at the start and come back to them at the end.

What you need

Item	How many	Used for
A4 scrap or copy paper	4 sheets per pair, plus 1 per student	Test shapes (3 per pair, 1 spare) and the record sheet
Books	2 per pair	The two sides of the bridge. Similar thickness.
Pebbles	About 15 per pair, plus 15 for you	The load. Roughly the same size. Black pebbles from the STEAM tub, or any small, equal weights (counters, blocks, coins).
Sticky tape	A short piece per pair	Closing the tube
Glue sticks	1 per pair	Gluing the record sheet into journals
STEAM journals	1 per student	Shelves by the door

How to do the activity

1. Set the gap

same gap for every test



Two books, one hand-span apart.

2. Make the shapes

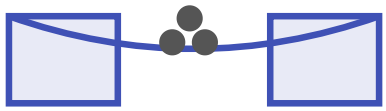
fold back and forth, or roll and tape



Fan folds run from book to book.

3. Add pebbles

one at a time, count each one



stop when it sags to the table

The last number is the score.

- **Gap:** place two books flat on the table, one hand-span apart. Keep the gap the same for every test so it is fair.
- **Flat:** lay a sheet across the gap like a bridge.
- **Fan:** fold a sheet back and forth like a paper fan. The folds must run from book to book, not across the gap.
- **Tube:** roll a sheet up and tape it closed.
- **Test:** place pebbles gently, one at a time, in the middle. Count each one. Stop when the paper sags down to the table or collapses. That number is the score.
- If pebbles roll off a tube, pile them on top, or count the pebble that falls as the last one.

Key words

strong, weak, shape, fold, roll, test, predict, weight

Lesson sequence

Time	Step	What to do and say
0 to 3 min	Settle	Students sit on the floor facing you. Read the Learning Intention and Success Criteria aloud.
3 to 10 min	Hook	Hold up one sheet of paper. Ask: *"Could this piece of paper hold up these pebbles?"* Thumbs up or down. Lay it flat across a gap between two books and place one pebble in the middle. It sags. Say: *"Paper is weak... or is it? Watch."* Fold a second sheet into a fan in front of them and lay it across the same gap. Add pebbles one by one with the class counting aloud. It holds! Ask: *"It is the same paper. What changed?"* (The shape. It was folded.) Do not show the tube yet.
10 to 14 min	Explain the challenge	Say: *"Today you are paper engineers. You will test 3 shapes: flat, fan and tube."* Roll a tube and tape it. Show how to place pebbles gently and when to stop. Ask each pair to point to the shape they predict will win. Draw the recording layout (below) on the board.

14 to 30 min	Test in pairs	Helpers hand out 2 books, about 15 pebbles, 4 sheets and a piece of tape per pair. One partner places pebbles while the other counts, then they swap for the next shape. Roam with the questions below.
30 to 38 min	Record	Give each student 1 sheet. They fold it in half, draw the 3 shapes on the front, write how many pebbles each held and circle the strongest. Helpers hand out journals. Students glue the sheet in and write the date and the title **Paper Power**.
38 to 45 min	Share and pack up	3 or 4 pairs share using the sentence stem on the board. Read the Success Criteria again and ask for thumbs. Finish with the key idea (below). Pack up: test paper into recycling, pebbles counted back into their container, books back, journals onto the shelves by the door.

Write this on the board

	Flat	Fan	Tube
Pebbles held	___ pebbles	___ pebbles	___ pebbles

Circle the strongest.

Sentence stem: **The _____ was strongest because _____.**

Questions to ask while roaming

- What happened when you added more pebbles?
- Is it the same paper? So what changed?
- Why do you think the fan held more than the flat paper?
- Which shape do you think will win next? Why?

Support and extension

Who	What to do
Support	Fold the first fan together with the pair. Students can record with drawings and numbers only. Pair a confident counter with a student who needs help counting.
Extension	Fast finishers (often Year 2) try a different fold, or a fat tube against a thin tube. Can they make a shape that holds even more? Write one sentence explaining why.

Key idea to finish on: Same paper, different shape, different strength. Folding and rolling made the paper stronger without adding anything new. Next week we will find out what happens when we twist paper together.

Notes for the teacher

- Any result is fine. Most pairs find the flat sheet holds almost nothing and the fan holds the most. The comparison is the learning, not the exact numbers.
- If a fan collapses straight away, check that the folds run from book to book.
- Pebbles stay on the table and are placed, not dropped. Count them back into their container at pack up.
- Nothing needs to be kept for next lesson.
- Please leave a short note about how far the class got and any students who were absent.

Curriculum links

- **Science VC2S2U05:** materials can be combined in a variety of ways for particular purposes; objects can have different properties from the materials they are made from.
- **Design and Technologies VC2TDE2C04:** explore the characteristics and properties of materials used to create designed solutions.
- **Maths:** counting a collection, comparing quantities.